

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460210.7516 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.016 +/- 0.046
Transit depth (Rp/Rs)^2	0.024 +/- 0.14 [%]
Orbital Inclination (inc)	89.81 +/- 1.68
Airmass coefficient 1 (a1)	1.006 +/- 0.024
Airmass coefficient 2 (a2)	-0.006 +/- 0.031
Scatter in the residuals of the lightcurve fit is	2.37 %
Best Comparison Star	#2 - [206, 451]
Optimal Aperture	3.3
Optimal Annulus	11.67
Transit Duration (day)	0.1 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.016 ± 0.046 vs 0.1418 (deviation 2.53σ)
Duration fit vs published	0.1 d vs 0.1196 d (deviation 1.01σ)
Mid-time O–C	0.4 min
Depth SNR	0.3
Residual scatter	2.37 %

Light curve

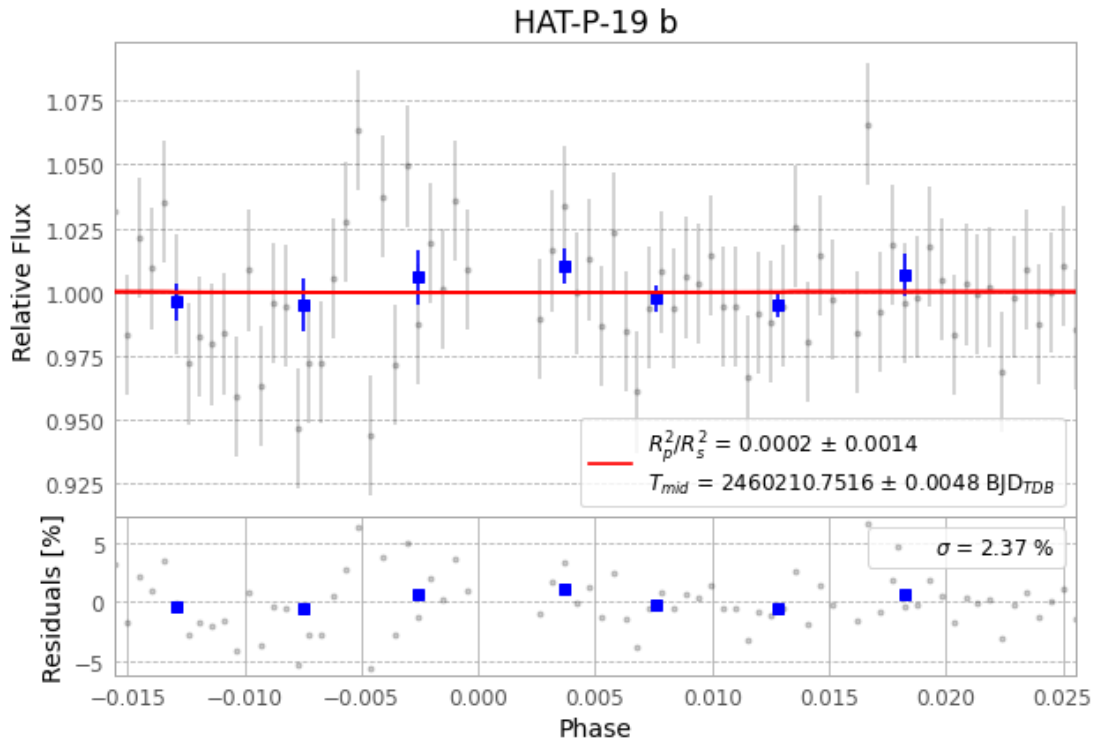


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-09-22.png (SHA-256 7201bbdd1ccb90b1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5ce22a2dcdd8e93a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

80 FITS frames (53 MB), HATP-19230923042415.FITS → HATP-19230923082115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-230923.log (6b866b7a7361...), FinalLightCurve_HAT-P-19 b_2023-09-22.png (7201bbdd1ccb...), FinalLightCurve_HAT-P-19 b_2023-09-22.csv (5d263a38a464...)

Compute resources

Wall time 295.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 02:17Z.